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RESEARCH ARTICLE

ALIVE: An Agentic Longitudinal IDE-IPA Value Evaluation Framework for Innovation-Driven Enterprise Assessment

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ABSTRACT Agentic artificial intelligence architectures have demonstrated transformative potential for automating complex, multi-dimensional evaluation tasks; however, existing deployments remain predominantly stateless, incapable of accumulating institutional knowledge across evaluation cycles and therefore structurally unfit for longitudinal domains such as Impact Pathway Assessment and Social Return on Investment scoring, which inherently demand temporal coherence and precedent-informed judgment. This article presents *ALIVE*: an Agentic Longitudinal IDE-IPA Value Evaluation Framework, a sixth-generation (L6) multi-agent architecture addressing this limitation through the PACT Loop: *Perceive* normalizes inputs and retrieves episodic priors from ChromaDB; *Analyze* scores 18 dimensions in parallel via a shared rubric loaded once across all agents; *Converge* applies an auto-calibrating halting rule terminating on score stability, target achievement, or budget exhaustion; and *Transfer* extracts lessons via idempotent atom derivation, matches peer patterns, and broadcasts learnings, with longitudinal state in PostgreSQL and ChromaDB ensuring each cycle improves the next. Four agents (Funder, Company, Researcher, and Society), each instantiated as a role-specific skill context, are orchestrated by Claude via 21 stateless MCP tools on Railway.app without server-side API key. A self-evaluation loop scores each response against a structured rubric, triggering retry until a quality-gate is satisfied. *ALIVE* operationalizes the IDE-IPA Analyzer-Pro V2.0, a 100-point, 18-dimension rubric spanning Standard IDE Assessment (Part A), Research-Specific Assessment (Part B), Impact Pathway Logic (Part C), and SROI Assessment (Part D). Validation on thirty synthetic proposals across six industry domains against a three-expert panel demonstrates $ICC_{21} = 0.922$ (≥ 0.80 good-agreement threshold), funding decision accuracy of 86.7%, and sub-five-minute processing per proposal, establishing *ALIVE* as a scalable, self-improving infrastructure for longitudinal impact pathway assessment in research funding administration.

INDEX TERMS Agentic artificial intelligence, impact pathway assessment, longitudinal evaluation, multi-agent systems, workflow orchestration.

I. INTRODUCTION

A. BACKGROUND AND MOTIVATION

Agentic artificial intelligence architectures have demonstrated transformative potential for automating complex, multi-dimensional evaluation tasks across scientific,

commercial, and policy domains [1], [2], [3]. The emergence of Large Language Model (LLM)-powered multi-agent systems, capable of autonomous goal-directed behavior, persistent memory management, and adaptive decision-making over extended task horizons, has opened new possibilities for systematizing evaluation processes that were previously dependent on scarce human expert panels [4], [5]. In particular, the evaluation of Innovation-Driven Enterprise (IDE)

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research proposals represents a domain where agentic AI offers substantial potential: proposals must be assessed across technological readiness, commercial viability, social impact pathways, and Social Return on Investment (SROI), requiring evaluators with simultaneously broad and deep expertise across multiple dimensions [6], [7].

National research funding systems increasingly mandate structured Impact Pathway Assessment (IPA) to justify public investment in science and technology [8]. The IPA logic model operationalizes causal pathways through sequential stages (Input, Activity, Output, Outcome, and Impact), each requiring evaluation for logical coherence and evidence support across extended project lifecycles [9], [10]. Critically, IPA and SROI assessment are inherently longitudinal processes: funding bodies must track proposal trajectories across multiple evaluation cycles, accumulate institutional knowledge about risk patterns, and apply precedent-informed judgment when assessing new submissions [11], [12]. This longitudinal requirement places structural demands on evaluation infrastructure that single-session, stateless AI pipelines are fundamentally unable to meet.

Despite growing adoption of AI-assisted evaluation tools, existing agentic AI deployments remain predominantly stateless: each evaluation cycle begins from a blank slate, discarding institutional knowledge accumulated in prior assessments [13], [14]. Human panel-based evaluation, while knowledgeable, suffers from well-documented limitations including evaluator fatigue, inconsistent rubric interpretation, cognitive bias, and scalability constraints when processing large proposal volumes [15], [16]. Inter-rater variability among human evaluators, even under identical rubric conditions, further undermines evaluation integrity and equity [17], [18]. These structural limitations collectively motivate the development of a new class of agentic AI architecture specifically designed for longitudinal, multi-dimensional evaluation.

B. PROBLEM STATEMENT

Four fundamental engineering gaps limit the current state of automated IDE-IPA evaluation. First, existing agentic AI systems operate without persistent episodic memory, being incapable of retrieving prior evaluation precedents or accumulating institutional risk knowledge across cycles, rendering them structurally incompatible with the longitudinal demands of IPA and SROI assessment. Second, current tools operate as single-model, single-dimension systems unable to simultaneously assess proposals across 18 dimensions and four assessment domains, whereas the heterogeneity of IDE-IPA evaluation demands specialized agents with domain-specific reasoning strategies [6], [7]. Third, existing orchestration frameworks have not examined MCP-based tool orchestration with role-specific skill context instantiation as a mechanism for agent specialization in heterogeneous rubric evaluation, leaving a structural orchestration gap unaddressed [13], [14]. Fourth, existing systems require continuous human monitoring to detect and correct evaluation

quality failures, with iteration limits hardcoded at deployment time rather than derived from convergence criteria over episodic execution history. These four gaps collectively establish the engineering motivation for ALIVE.

C. CONTRIBUTIONS OF THE WORK

To address these four gaps, this paper presents ALIVE: an Agentic Longitudinal IDE-IPA Value Evaluation Framework, a sixth-generation (L6) multi-agent architecture that makes the following six contributions:

- 1) **PACT Loop: A Four-Phase Longitudinal Engine.** A structured evaluation pipeline comprising Perceive, Analyze, Converge, and Transfer phases, in which longitudinal state persisted across PostgreSQL and ChromaDB ensures each evaluation cycle retrieves episodic priors and improves subsequent ones, addressing a gap in existing agentic evaluation frameworks that lack explicit support for longitudinal IPA and SROI assessment.
- 2) **Four Stakeholder-Role Agents via MCP Orchestration.** Four agents (Funder, Company, Researcher, and Society), instantiated as role-specific skill contexts and orchestrated by Claude via 21 MCP tools hosted on Railway.app without server-side API key, supporting parallel dimension scoring within phases and sequential agent handoff across assessment roles.
- 3) **IDE-IPA Analyzer-Pro V2.0.** A 100-point, 18-dimension rubric across four assessment domains: Standard IDE Assessment (Part A, 28 pts), Research-Specific Assessment (Part B, 22 pts), Impact Pathway Logic (Part C, 25 pts), and SROI Assessment (Part D, 25 pts).
- 4) **Self-Evaluation Loop with Quality-Gate Enforcement.** A rubric-based output scoring mechanism that evaluates each agent response against a structured checklist and triggers automated retry until a weighted quality-gate threshold is satisfied, providing bounded quality assurance without continuous human monitoring.
- 5) **Idempotent Cross-Portfolio Knowledge Transfer.** A Knowledge Broker architecture implementing deterministic lesson atom derivation via MD5-based idempotent identifiers, peer pattern matching via ChromaDB semantic search, and priority-based broadcast to peer portfolios, enabling institutional knowledge to accumulate and propagate across evaluation cycles without duplication.
- 6) **Empirical Validation.** A 30-proposal synthetic corpus across six industry domains (AI & Data Science, AgriTech & Biotechnology, EV & Clean-Tech, Semiconductor & Advanced Manufacturing, HealthTech & Medical Devices, and FoodTech & BCG Economy), evaluated against three domain experts under three experimental conditions, demonstrating $ICC_{21} = 0.922 (\geq 0.80)$ and funding decision accuracy of 86.7%.

This paper evaluates ALIVE across four assessment domains in an ex-ante context using a synthetic proposal corpus. The remainder of this paper is organized as follows. Section II reviews related work across five thematic areas. Section III presents the ALIVE framework architecture, the PACT Loop algorithm, and the IDE-IPA Analyzer-Pro V2.0 rubric. Section IV describes the experimental design, synthetic proposal corpus, and evaluation protocol. Section V reports empirical results across four metrics (M1–M4) under three experimental conditions (C1–C3). Section VI discusses key findings, limitations, and future research directions. Section VII presents the conclusions.

II. LITERATURE REVIEW

A. AGENTIC AI AND AUTONOMOUS ENTERPRISE SYSTEMS

Agentic AI refers to artificial intelligence systems capable of autonomous goal-directed behavior, tool use, persistent memory management, and adaptive decision-making over extended task horizons [19], [20]. Unlike conventional AI systems optimized for narrow, constrained tasks, agentic architectures maintain internal state, decompose complex objectives into sequential sub-tasks, and adapt their strategies based on environmental feedback [21], [22]. A comprehensive survey of LLM-based agentic systems identifies five foundational components: agent profiling, environmental perception, self-directed action, mutual inter-agent interaction, and evolutionary adaptation, with persistent memory storage as the critical knowledge infrastructure enabling cross-session learning [2]. The specialization principle, which assigns distinct agents to well-defined subtasks, improves both output quality and interpretability in complex reasoning tasks [5], [13].

The application of agentic AI to enterprise decision-making has accelerated substantially, spanning domains from e-commerce automation [23] and ERP decision support [24] to clinical documentation review [25] and IT autonomy [26]. In each domain, the architectural shift from reactive to agentic AI yields measurable gains in task completion quality, process consistency, and operational scalability [4], [21]. However, existing enterprise deployments remain predominantly single-session: agents execute tasks within bounded contexts without accumulating institutional knowledge across operational cycles [13], [14]. This statelessness fundamentally limits their applicability to domains requiring longitudinal reasoning, precedent-informed judgment, and cross-cycle knowledge transfer, which are precisely the characteristics demanded by Impact Pathway Assessment and SROI evaluation [27].

B. WORKFLOW ORCHESTRATION FOR MULTI-AGENT PIPELINES

Workflow orchestration in LLM-based multi-agent systems encompasses content classification and agent assignment, task lifecycle management, inter-agent communication

protocols, and output aggregation from heterogeneous agent responses. Static orchestration frameworks based on pre-defined directed acyclic graphs lack the adaptability required for heterogeneous proposal evaluation, where proposal content, domain complexity, and dimensional emphasis vary substantially across submissions. Dynamic frameworks supporting online task discovery, adaptive replanning, and agent coordination based on runtime content analysis are better suited to variable-input evaluation demands [1], [4], [13].

The Model Context Protocol (MCP) has emerged as a standardized transport layer for exposing tool capabilities to LLM-based orchestrators, enabling server-side tool execution to be decoupled from client-side reasoning. Under this architecture, the LLM acts as the orchestrator, selecting and invoking tools based on task context, while the server exposes stateless functional endpoints without embedding routing or dispatch logic. This separation cleanly supports hybrid execution patterns: parallel invocation of independent scoring tools within a single assessment phase, and sequential agent handoff across phases where downstream agents depend on upstream outputs [1], [4]. Role-specific skill contexts, loaded as structured prompts rather than instantiated as separate processes, provide an efficient mechanism for agent specialization without the operational overhead of multi-process coordination [3]. These orchestration principles directly inform the ALIVE design, in which Claude orchestrates 21 MCP tools hosted on Railway.app across the four PACT Loop phases. Convergence analysis in multi-agent systems has been examined in related contexts [37], [38].

C. LLM-BASED AUTOMATED EVALUATION AND HALLUCINATION MITIGATION

The use of LLMs as automated evaluators, commonly termed “LLM-as-a-judge,” has gained traction across academic assessment, clinical documentation review, and code quality evaluation. Structured evaluation tasks place particular demands on LLM reliability: outputs must conform to predefined scoring schemas, remain within dimensional boundaries, and maintain logical consistency across interdependent assessment criteria. Hallucination in structured evaluation manifests as boundary violations where scores exceed rubric maxima, fabricated evidence citations unsupported by proposal content, and inter-dimensional contradictions violating rubric logical constraints [13], [25].

Structured output enforcement via JSON schema constraints represents an effective first-line hallucination mitigation strategy, eliminating a large class of format violations before downstream processing. Cross-agent verification, in which a secondary agent audits primary evaluator outputs for logical consistency, provides more robust hallucination coverage than single-agent self-checking alone, particularly for inter-dimensional contradiction detection [3], [13]. A bounded retry architecture with maximum iteration limits and quality-gate escalation prevents infinite loops while

TABLE 1. Research gap summary and ALIVE contributions relative to existing literature (✓ = addressed | ∅ = partially addressed | — = not addressed).

Gap / Theme	Limitation in existing work	Rep. studies	Agentic AI eval.	Orch. frmwks	IPA/ SROI	ALIVE contribution
G1 — Stateless arch.	No episodic memory; institutional knowledge lost between sessions	[2], [3], [13], [14]	—	—	—	PACT Loop with ChromaDB episodic KB + PostgreSQL persistent state; each cycle retrieves priors and improves the next
G2 — Single-model eval.	Unable to assess 18 dims & 4 domains; no stakeholder-role specialization	[3], [13]	—	∅	—	4 agents (Funder, Company, Researcher, Society) as role-specific skill contexts; parallel dim. scoring via shared rubric loaded once
G3 — MCP tool orch.	No framework applies MCP-based dispatch with role-specific skill context for rubric evaluation	[1], [4]	—	—	—	Claude orchestrates 21 MCP tools on Railway.app; hybrid parallel scoring + sequential agent handoff
G4 — Output quality	Mitigation layers studied in isolation; no quality-gate loop with empirical validation	[13], [25], [28]	∅	—	—	Self-evaluation loop with rubric checklist; automated retry until quality-gate threshold satisfied
G5 — No auto halting	Iteration limits hardcoded; no self-calibrating convergence criterion	[13], [14]	—	—	—	Auto-calibrating halting rule (Alg. 1): C1 score stability, C2 target met, C3 budget exhaustion; guaranteed termination
G6 — IPA/SROI auto.	Largely manual; no LLM framework covering all 5 IPA stages and 3 SROI scenarios	[9]–[12]	—	—	∅	IDE-IPA Analyzer-Pro V2.0: 100-pt 18-dim rubric; Parts A–D automated; 3-scenario SROI; idempotent lesson atoms

preserving evaluation completeness. Human-centered design maintaining oversight and override capability is consistently identified as essential for institutional adoption of automated evaluation systems [25], [28]. ALIVE addresses output reliability through a self-evaluation loop that scores each agent response against a structured rubric checklist and triggers automated retry until a weighted quality-gate threshold is satisfied, providing bounded quality assurance without continuous human monitoring.

D. IMPACT PATHWAY ASSESSMENT AND MULTI-DIMENSIONAL READINESS

Impact Pathway Assessment provides a structured methodology for evaluating the anticipated effects of research projects through systematic causal pathway analysis [10], [11]. The IPA logic model operationalizes causal pathways through five sequential stages (Input, Activity, Output, Outcome, and Impact), each evaluated for logical coherence, evidence support, and temporal plausibility [9]. Multi-dimensional readiness assessment frameworks extend beyond Technology Readiness Level (TRL) to encompass commercial, social, environmental, and policy readiness dimensions, reflecting the increasing complexity of innovation ecosystems that funding bodies must evaluate [29]. The integration of SROI modeling into readiness assessment adds a quantitative economic dimension, requiring evaluators to project discounted social value streams against investment inputs using standardized financial proxies [10], [12].

Ex-ante evaluation, defined as assessment conducted prior to project implementation based on projected rather than realized outcomes, presents particular methodological challenges

for automated systems, as evaluators must reason about counterfactual impact trajectories under uncertainty [10], [11]. National research funding systems increasingly mandate multi-dimensional IPA evaluation, requiring proposals to demonstrate alignment across technological, commercial, social, and national strategy dimensions simultaneously [6], [8]. The longitudinal nature of IPA, in which evaluation quality depends critically on accumulated precedent knowledge from prior assessment cycles, distinguishes it from single-session evaluation domains and motivates the architectural design choices of ALIVE [9], [12]. AI-based Decision Support Systems have demonstrated improvements in decision quality and assessment consistency across complex multi-dimensional evaluation tasks [24], [28], [30].

E. LIMITATIONS OF EXISTING STUDIES AND RESEARCH GAP

Despite substantial progress across the four themes reviewed above, a consistent pattern of limitations emerges that collectively motivate the ALIVE framework. Existing agentic AI evaluation systems have been assessed predominantly on task completion accuracy metrics, without rigorous treatment of longitudinal knowledge accumulation, cross-session routing improvement, or institutional memory formation [13], [14]. No existing framework explicitly addresses the structural incompatibility between stateless LLM pipelines and the longitudinal demands of IPA and SROI assessment, leaving a fundamental architectural gap unaddressed [2], [3]. Multi-agent orchestration research has demonstrated dynamic routing superiority over static pipelines but has not examined MCP-based tool orchestration with role-specific skill context

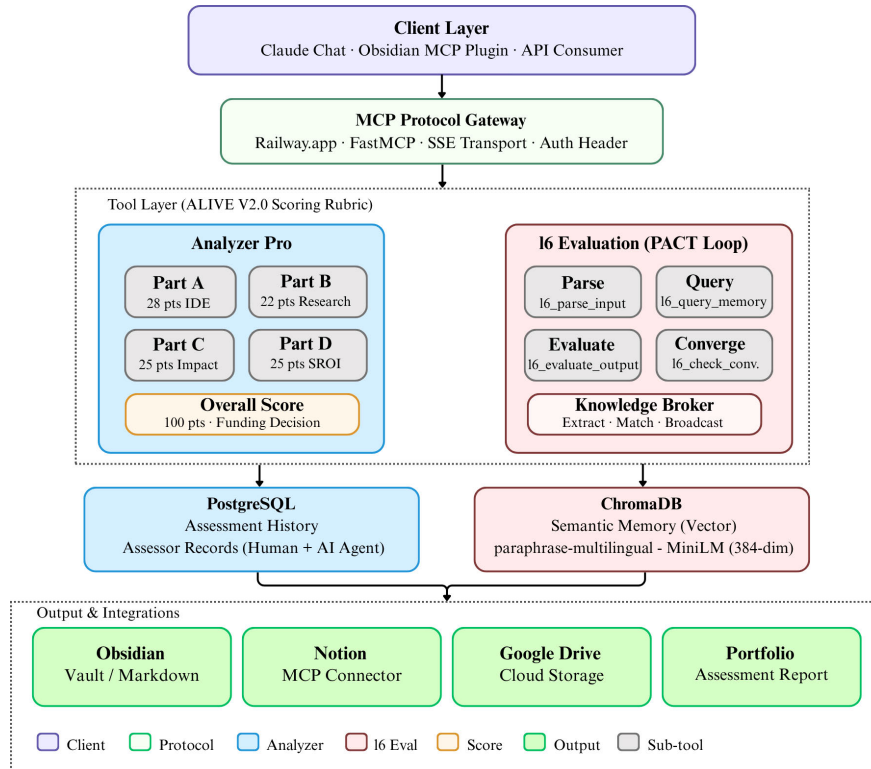


FIGURE 1. Overall ALIVE system architecture showing the layered client-protocol-tool-storage design with Claude as orchestrator, 21 stateless MCP tools (analyzer pro and L6 Evaluation groups), PostgreSQL and ChromaDB persistent backends, and four output integration endpoints.

instantiation as a mechanism for agent specialization in heterogeneous rubric evaluation [1], [4].

Hallucination mitigation strategies in the literature address individual mitigation layers in isolation (schema enforcement, cross-agent verification, or retry logic) without integrating them into a unified quality-gate architecture with empirical validation in structured scoring contexts [13], [25]. Episodic memory and retrieval-augmented generation have been examined independently in the literature, without demonstrating combined shared-rubric and episodic-retrieval architectures that eliminate per-agent rubric retransmission while constraining context to relevant precedents [2], [13]. IPA and SROI assessment automation remains largely manual, with no published framework demonstrating LLM-based evaluation across all five IPA stages and three SROI scenarios simultaneously [10], [11], [12]. Collectively, these limitations distinguish ALIVE from prior multi-agent and longitudinal evaluation systems, which have not simultaneously addressed stateless architectural constraints, MCP-based tool orchestration with role-specific skill context instantiation, and cross-cycle idempotent knowledge transfer within a unified evaluation framework. Table 1 synthesizes these gaps, positioning ALIVE’s contributions relative to the existing literature.

III. METHODOLOGY

This section describes the design, architecture, and algorithms of ALIVE: Agentic Longitudinal IDE-IPA Value

Evaluation Framework. ALIVE is a sixth-generation (L6) multi-agent system orchestrated by Claude via a Model Context Protocol (MCP) server deployed on Railway.app. The methodology is organized into six subsections: (A) system architecture, (B) workflow integration, (C) the PACT Loop algorithm, (D) the IDE-IPA Analyzer-Pro rubric, (E) result storage and knowledge-base sharing, and (F) implementation and scalability considerations.

A. ALIVE ARCHITECTURE

The overall ALIVE system follows a layered client-protocol-tool-storage architecture, as illustrated in Fig. 1. At the topmost layer, Claude (claude.ai Max Plan) acts as the AI orchestrator. Rather than embedding intelligence inside server-side logic, all reasoning, planning, and decision-making reside in the client-side language model. This design is intentional: it eliminates server-side API-key management and keeps proprietary model weights off the deployment infrastructure [31].

Claude communicates downward through a secure **MCP Protocol Gateway** hosted on Railway.app using FastMCP with Server-Sent Events (SSE) transport and Bearer-token authentication. Below this gateway, ALIVE exposes **21 stateless MCP tools** organized into two functional groups (Table 2): (i) Analyzer Pro (nine tools) that implement the IDE-IPA Analyzer-Pro V2.0 scoring rubric, and (ii) L6 Evaluation (twelve tools) that implement the PACT Loop,

TABLE 2. ALIVE MCP tool registry.

Tool Name	Group	Function
<i>Analyzer Pro — 9 tools (IDE-IPA V2.0 scoring rubric)</i>		
<code>analyzer_pro_load_framework</code>	Analyzer Pro	Load shared IDE-IPA V2.0 rubric into session context
<code>analyzer_pro_score_part_a/b/c/d</code>	Analyzer Pro	Score Parts A–D independently (parallel execution)
<code>analyzer_pro_overall_score</code>	Analyzer Pro	Aggregate raw scores; map to funding decision tier
<code>analyzer_pro_calculate_sroi</code>	Analyzer Pro	Compute three-scenario SROI with deadweight, attribution, drop-off
<code>analyzer_pro_log_coaching_session</code>	Analyzer Pro	Persist coaching notes to PostgreSQL COACHING_SESSIONS
<code>analyzer_pro_get_history</code>	Analyzer Pro	Retrieve longitudinal assessment history for a company
<i>L6 Eval — 12 tools</i>		
<code>l6_parse_input</code>	L6 Eval	Normalise heterogeneous inputs to structured JSON (Phase P)
<code>l6_query_memory</code>	L6 Eval	Semantic search ChromaDB for episodic context (Phase P)
<code>l6_evaluate_output</code>	L6 Eval	Keyword-weighted rubric self-check; return pass/fail + feedback (Phase C)
<code>l6_check_convergence</code>	L6 Eval	Apply C1/C2/C3 halting rule to score history S (Phase C)
<code>l6_write_shared_kb</code>	L6 KB	Embed finding and upsert to ChromaDB <i>shared_kb</i> (Phase T)
<code>l6_read_shared_kb</code>	L6 KB	Cosine-similarity retrieval from <i>shared_kb</i> (any agent)
<code>l6_kb_extract_lessons</code>	L6 KB	Derive idempotent lesson atoms via MD5 <i>atom_id</i> (Phase T)
<code>l6_kb_match_peers</code>	L6 KB	Find top- k peer companies by embedding similarity (Phase T)
<code>l6_kb_broadcast_lessons</code>	L6 KB	Propagate lessons to role-specific audiences (Phase T)
<code>l6_kb_flag_critical_risk</code>	L6 KB	Escalate critical risk; auto-set deadline by severity (Phase T)
<code>l6_reset_shared_kb</code>	L6 Utility	Delete and recreate <i>shared_kb</i> ; re-embed via <i>paraphrase-multilingual-MiniLM-L12-v2</i>
<code>l6_chroma_status</code>	L6 Utility	Report ChromaDB status: embedding model, collection names, document counts

self-evaluation, convergence checking, and Knowledge Broker dissemination.

At the storage layer, ALIVE maintains two persistent backends that collectively provide the longitudinal memory distinguishing it from stateless evaluation systems. **PostgreSQL** stores structured relational data, including assessment records, part-level dimension scores, coaching-session logs, and convergence traces, enabling temporal queries across multiple evaluation cycles. **ChromaDB** stores semantic vector memory using the *paraphrase-multilingual-MiniLM-L12-v2* embedding model (384 dimensions). ChromaDB houses the shared knowledge base (*shared_kb*), agent-specific episodic collections, idempotent lesson atoms, risk alerts, and broadcast records. Together, these two stores ensure that knowledge accumulated during one assessment cycle is available to all subsequent cycles across the full portfolio.

Four role-specific agent contexts, namely A1 (Company), A2 (Society), A3 (Researcher), and A4 (Funder), are implemented as *SKILL.md* context files rather than separate processes. The Company agent (A1) adopts the perspective of the IDE candidate, assessing business model viability, innovation capability, and market readiness. The Society agent (A2) reflects the Intermediary perspective, evaluating social impact pathways, SROI projections, and community benefit. The Researcher agent (A3) instantiates the role of an Innovation Business Development Services specialist, scoring research quality, technology transfer potential, and scalability. The Funder agent (A4) evaluates funding risk, return on investment, and national research and innovation

policy alignment. Claude loads the appropriate context file at runtime according to the stakeholder role being served, enabling the same 21 MCP tools to serve all four assessment perspectives without additional deployment overhead.

B. WORKFLOW INTEGRATION

ALIVE integrates into the IDE-IPA administrative workflow through a structured pre-assessment, assessment, and post-assessment pipeline, depicted in Fig. 2. Prior to formal scoring, the system verifies company registration and legal status, collects the comprehensive project proposal, gathers baseline assessment data, and reviews supporting documents. This gatekeeping phase ensures input quality before computational resources are allocated.

During the Assessment Phase, Parts A through D are scored in parallel by independent MCP tool calls, each operating on the same structured input context. Score independence is critical: because no part shares state with another during scoring, the system avoids cascading bias that would arise if an early partial score influenced subsequent dimensions. After all four part scores are returned, they are aggregated by `analyzer_pro_overall_score` into a 100-point total, and the Funding Decision Logic maps the aggregate into one of four tiers as follows:

- Overall score ≥ 85 , all thresholds met: **Tier 1 STRONGLY FUND** (full funding, proceed to contract).
- Overall score ≥ 64 , all thresholds met: **Tier 2 FUND** (full funding, monitor closely).

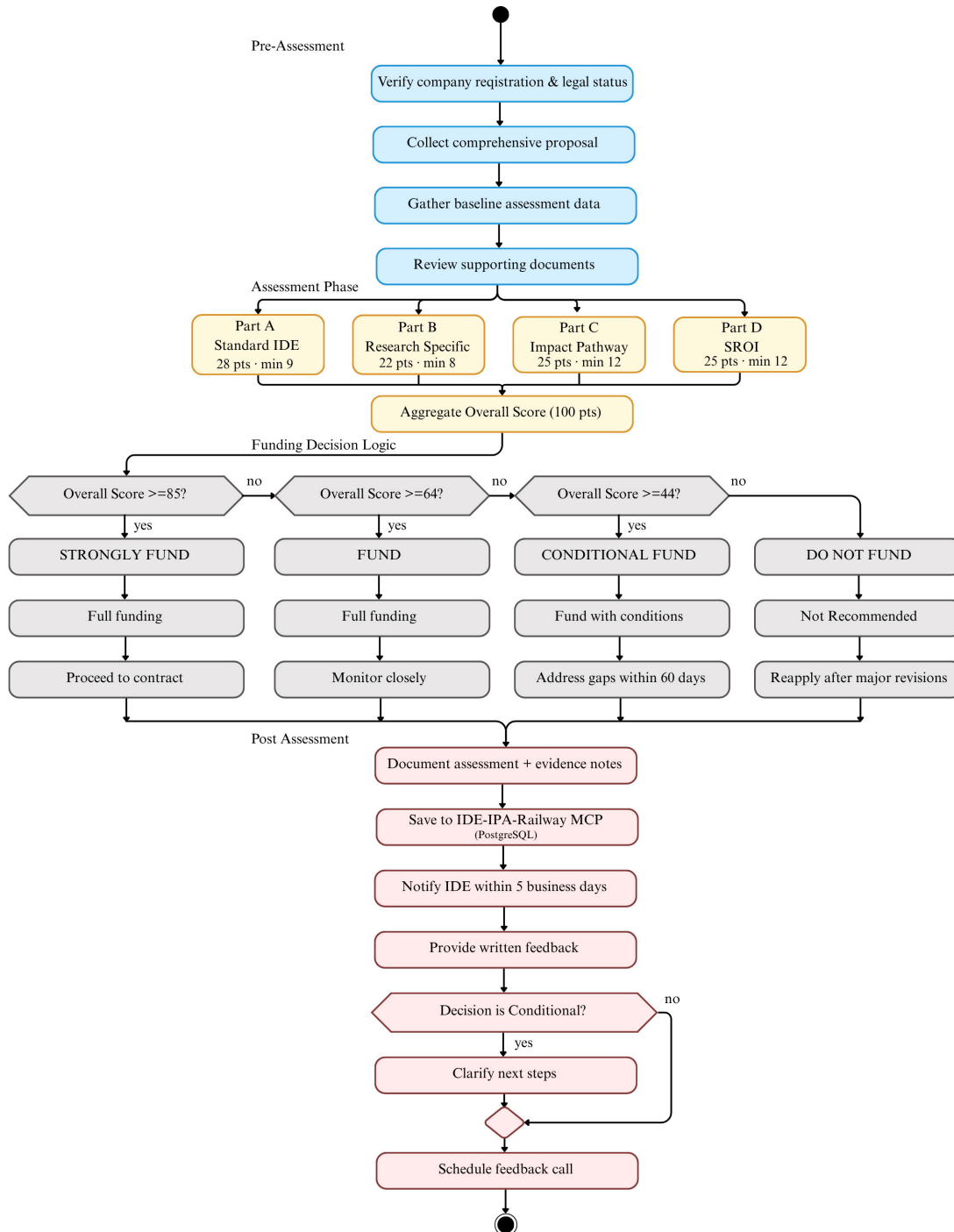


FIGURE 2. ALIVE-Integrated IDE-IPA administrative workflow comprising pre-assessment gatekeeping, parallel four-part scoring with threshold-based funding decision logic, and post-assessment coaching cycle initiation.

- Overall score ≥ 44 , all thresholds met: **CONDITIONAL FUND** (fund with conditions; address gaps within 60 days).
- Overall score < 44 or any threshold fails: **DO NOT FUND** (not recommended; reapply after major revisions).

All four parts must individually meet their minimum thresholds (Part A ≥ 9 , Part B ≥ 8 , Part C ≥ 12 , Part D ≥ 12)

regardless of total score. A company that fails any part-level threshold receives a **DO NOT FUND** decision immediately, as these thresholds represent non-negotiable quality gates rather than weighted criteria that can be compensated by performance in other dimensions.

The Post-Assessment Phase commits all scoring outputs and evidence notes to the MCP server, notifies the IDE program office within five business days, provides written

feedback to the applicant, and, for conditional decisions, schedules a feedback call and clarifies remediation steps. This structured handoff closes the assessment loop and initiates the coaching cycle if required.

Agent coordination follows a **hybrid parallel-sequential orchestration model**. Parts A, B, C, and D are executed in parallel (they are independent functions), while the agent pipeline $A1 \rightarrow A2 \rightarrow A3 \rightarrow A4$ is sequential, with each agent consuming the outputs of the prior stage before generating its own deliverables. Claude acts as the orchestrator throughout, selecting and invoking tools dynamically without a separate FastAPI dispatcher layer.

C. PROPOSED ALGORITHM: THE PACT LOOP

The core computational mechanism of ALIVE is the **PACT Loop** depicted in Fig. 3, a four-phase agentic evaluation cycle: *Perceive*, *Analyze*, *Converge*, and *Transfer*. The loop is parameterized by a convergence tolerance $\varepsilon = 0.02$, a quality target $\tau = 0.85$, and a coaching budget $N_{\max} = 5$ iterations.

Phase P — Perceive. The loop begins by normalizing the raw proposal ρ (which may be PDF, text, Notion export, or image) into a structured JSON object x_0 via `l6_parse_input`. Simultaneously, `l6_query_memory` performs a semantic search over the company-specific ChromaDB collection to retrieve episodic context ε from prior assessment cycles. This retrieval step is the primary mechanism through which historical knowledge informs the current evaluation, operationalizing longitudinal coherence from the first iteration.

Phase A — Analyze. The Analyze phase executes the four-part rubric scoring in a repeat loop. On each iteration t , `ScorePartA(c, xt)`, `ScorePartB(c, xt)`, `ScorePartC(c, xt)`, and `ScorePartD(c, xt)` return dimension scores a_t, b_t, c_t, d_t within their respective ranges $[0, 28]$, $[0, 22]$, $[0, 25]$, $[0, 25]$. The normalized composite score is computed as:

$$s_t = \frac{a_t + b_t + c_t + d_t}{100} \quad (1)$$

and appended to the score history S and full assessment trace T . A self-evaluation subroutine, `l6_evaluate_output`, then applies a keyword-weighted quality rubric with threshold 0.85 to the generated assessment text, returning a weighted score, pass/fail verdict, and dimension-level gap feedback. When the quality gate is not satisfied, Claude interprets the returned feedback and may invoke additional scoring iterations before advancing to convergence checking.

Phase C — Converge. After each scored iteration, `l6_check_convergence` evaluates three halting conditions against the score history S , as formalized in Table 3. The function computes the absolute score delta:

$$\Delta = |S[t] - S[t - 1]| \quad (2)$$

and applies conditions C1, C2, and C3. Note that the absolute-value formulation of Δ renders the halting rule non-monotonic tolerant: it halts on score stability regardless of whether the score trajectory is ascending or descending,

which is important for proposals that degrade under adversarial coaching. If no condition is satisfied, a `CoachingUpdate` is applied to generate x_{t+1} from rubric gap feedback, and the loop increments t . The full `L6_CheckConvergence` algorithm is given in Algorithm 1.

Upon convergence, the funding decision d is derived by `DecisionMap(st*, rt)`, mapping the final normalized score and halting reason to one of {STRONGLY FUND, FUND, CONDITIONAL FUND, DO NOT FUND} according to the thresholds defined in Section III-B. This mapping is decidable in $O(1)$ time as the decision space comprises a finite partition of the score interval $[0, 1]$ crossed with a finite set of four halting reasons.

Algorithm 1 implements an auto-calibrating halting rule that terminates the coaching loop based on observed score behavior rather than a fixed iteration count. At each iteration t , the absolute score delta $\Delta = |s_t - s_{t-1}|$ is computed and evaluated against three conditions: C1 is satisfied when $\Delta < \varepsilon = 0.02$, indicating score stabilization regardless of whether the trajectory is ascending or descending (non-monotonic tolerant); C2 is satisfied when $s_t \geq \tau = 0.85$ at any iteration, enabling early exit for high-quality proposals; and C3 is satisfied when $t \geq N_{\max} - 1 = 4$, providing a hard budget bound. The conditions are evaluated sequentially, with C1 taking precedence over C2 when score stability and target achievement coincide.

Phase T — Transfer. Upon convergence, the Transfer phase disseminates findings across the shared knowledge ecosystem through five sequential Knowledge Broker operations (Fig. 4):

- `l6_write_shared_kb`: persists assessment findings, tags, and 384-dimensional embedding to the ChromaDB *shared_kb* collection.
- `l6_kb_match_peers`: performs cosine-similarity search (Eq. 3) to identify the top- k peer companies sharing similar assessment profiles.
- `l6_kb_extract_lessons`: derives idempotent lesson atoms from coaching session content using MD5-based deterministic atom identifiers, preventing duplicate knowledge accumulation.
- `l6_kb_flag_critical_risk`: escalates risk alerts with severity-calibrated deadlines and propagates lessons to role-specific stakeholder audiences.
- `l6_kb_broadcast_lessons`: propagates lessons to role-specific stakeholder audiences.

The peer-similarity score used in `l6_kb_match_peers` is computed as:

$$\text{similarity}(A, B) = \frac{A \cdot B}{\|A\| \cdot \|B\|} \quad (3)$$

where A and B are 384-dimensional sentence embeddings of the query and candidate documents, respectively. Documents with similarity below a minimum relevance threshold (default 0.35) are filtered before grouping by agent type,

TABLE 3. PACT loop convergence conditions ($\epsilon = 0.02$, $\tau = 0.85$, $N_{\max} = 5$).

Condition	Trigger	Reason String	Semantics
C1	$\Delta < \epsilon$ AND $s_t < \tau$	converged_below_target	Score stabilised but below target
C1*	$\Delta < \epsilon$ AND $s_t \geq \tau$	converged_target_met	Score stabilised at or above target
C2	$s_t \geq \tau$ (any iteration)	target_met_early	Quality gate satisfied proactively
C3	$t \geq N_{\max} - 1$	max_iter_reached	Coaching budget exhausted

C1* denotes the converged-and-target-met case, which is the preferred termination state.

Algorithm 1 L6_CheckConvergence — Auto-calibrating Halting Rule for Iterative Assessment

Require: Company identifier c ; score history $S = (s_0, s_1, \dots, s_t)$ with $s_t \in [0, 1]$; tolerance $\epsilon = 0.02$; target threshold $\tau = 0.85$; iteration budget $N_{\max} = 5$

Ensure: Tuple $\langle \text{halt}, \text{reason}, t \rangle$ where $\text{halt} \in \{\text{true}, \text{false}\}$

```
1:  $t \leftarrow |S| - 1$                                 ▷ Current iteration index
2: if  $t = 0$  then
3:   return  $\langle \text{false}, "insufficient\_history", 0 \rangle$ 
4: end if
5:  $\Delta \leftarrow |s_t - s_{t-1}|$                         ▷ Absolute score change (non-monotonic tolerant)
6: ▷ Condition C1: Convergence — score change below tolerance
7: if  $\Delta < \epsilon$  then
8:   if  $s_t \geq \tau$  then
9:     return  $\langle \text{true}, "converged\_target\_met", t \rangle$                                 ▷ Optimal: fund
10:  else
11:    return  $\langle \text{true}, "converged\_below\_target", t \rangle$                                 ▷ Structural gap: reject
12:  end if
13: end if
14: ▷ Condition C2: Target achievement — early success
15: if  $s_t \geq \tau$  then
16:   return  $\langle \text{true}, "target\_met\_early", t \rangle$                                 ▷ Fund: early target
17: end if
18: ▷ Condition C3: Budget exhaustion — bounded rationality
19: if  $t \geq N_{\max} - 1$  then
20:   return  $\langle \text{true}, "max\_iter\_reached", t \rangle$                                 ▷ Trend-dependent decision
21: end if
22: return  $\langle \text{false}, "in\_progress", t \rangle$                                 ▷ Continue iterating
```

TABLE 4. IDE-IPA analyzer-pro V2.0: four-part rubric structure.

Part	Domain	Max	Min	Dims	Key Dimensions
A	Standard IDE Assessment	28	9	6	Dim1: Innovation Capability (6 pts); Dim2: Business Model (6 pts); Dim3: Team & Organization (4 pts); Dim4: Market Readiness (4 pts); Dim5: Financial Sustainability (4 pts); Dim6: Network & Partnerships (4 pts)
B	Research-Specific Assessment	22	8	4	B7: Research Quality (8 pts); B8: Technology Transfer (6 pts); B9: Scalability & Replication (6 pts); B10: National Strategy Alignment (2 pts)
C	Impact Pathway Logic (Ex-Ante)	25	12	5	C1: Pathway Architecture (7 pts); C2: Causal Logic (8 pts); C3: Assumptions & Risk (5 pts); C4: Impact Projection (3 pts); C5: Adaptive Management (2 pts)
D	SROI Assessment	25	12	3	D1: Forecast SROI Model (10 pts); D2: SROI Calculation (8 pts); D3: Evaluative SROI Plan (7 pts)
Total	IDE-IPA Analyzer-Pro V2.0	100	All pass	18	All four thresholds must be satisfied simultaneously for overall approval

Algorithm 2 PACT_Loop — ALIVE Agentic Longitudinal Assessment Cycle

Require: Raw proposal ρ (text, PDF, Notion, or image); company identifier c ; rubric $\mathcal{R}$ (ALIVE V2.0: Parts A–D, max = {28, 22, 25, 25}); coaching budget $N_{\max} = 5$; target $\tau = 0.85$; tolerance $\varepsilon = 0.02$; shared knowledge base $\mathcal{K}$ (ChromaDB)

Ensure: Assessment trace $\mathcal{T} = \{(s_t, x_t, r_t)\}_{t=0}^T$; funding decision $d \in \{\text{Strongly Fund, Fund, Conditional Fund, Do Not Fund}\}$; lessons $\mathcal{L}$ broadcast to peer portfolio

- 1: $\triangleright$ Phase 0 — Initialization
- 2: $S \leftarrow []$; $\mathcal{T} \leftarrow []$; $t \leftarrow 0$
- 3: $\triangleright$ Phase P (Perceive) — Unified Input Normalization
- 4: $x_0 \leftarrow \text{L6_ParseInput}(\rho, \text{type}(\rho))$ $\triangleright \rightarrow$ structured JSON
- 5: retrieve episodic context: $\mathcal{E} \leftarrow \text{L6_QueryMemory}(c, x_0, \text{task} = \text{"ide_assessment"})$
- 6: $\triangleright$ Phase A (Analyze) — Four-Part Assessment Loop
- 7: **repeat**
- 8: $a_t \leftarrow \text{ScorePartA}(c, x_t)$ $\triangleright a_t \in [0, 28]$: IDE Standard
- 9: $b_t \leftarrow \text{ScorePartB}(c, x_t)$ $\triangleright b_t \in [0, 22]$: Research-Specific
- 10: $c_t \leftarrow \text{ScorePartC}(c, x_t)$ $\triangleright c_t \in [0, 25]$: Impact Pathway
- 11: $d_t \leftarrow \text{ScorePartD}(c, x_t)$ $\triangleright d_t \in [0, 25]$: SROI
- 12: $\sigma_t \leftarrow a_t + b_t + c_t + d_t$; $s_t \leftarrow \sigma_t/100$
- 13: append s_t to S ; append $(s_t, x_t, a_t, b_t, c_t, d_t)$ to $\mathcal{T}$
- 14: $\triangleright$ Phase C (Converge) — Self-Evaluation Check (Algorithm 1)
- 15: $\langle h_t, r_t, t \rangle \leftarrow \text{L6_CheckConvergence}(c, S, \varepsilon, \tau, N_{\max})$
- 16: **if** $h_t = \text{true}$ **then**
- 17: **break** $\triangleright$ Halting criterion satisfied
- 18: **end if**
- 19: $\triangleright$ Coaching intervention: update x_{t+1} via rubric feedback
- 20: $x_{t+1} \leftarrow \text{CoachingUpdate}(x_t, \mathcal{R}, \text{gaps}(a_t, b_t, c_t, d_t))$
- 21: $\text{LogCoachingSession}(c, t + 1, \text{focus}, s_{t-1}, s_t)$
- 22: $t \leftarrow t + 1$
- 23: **until** $t \geq N_{\max}$
- 24: $\triangleright$ Funding Decision Mapping
- 25: $d \leftarrow \text{DecisionMap}(s_t, r_t)$ $\triangleright$ See Table 3
- 26: $\triangleright$ Phase T (Transfer) — Knowledge Broker Dissemination
- 27: $\text{L6_WriteSharedKB}(c, \text{finding}(s_t, r_t, d), \text{tags}(x_t))$ $\triangleright$ Persist to $\mathcal{K}$
- 28: $\mathcal{P} \leftarrow \text{L6_KBMatchPeers}(\text{query}(x_t), \text{topK}=3)$ $\triangleright$ Cross-portfolio peer discovery
- 29: $\mathcal{L} \leftarrow \text{L6_KBExtractLessons}(\text{query}(c, d))$ $\triangleright$ Atomize transferable lessons
- 30: **if** risk flags detected in x_t **then**
- 31: $\text{L6_KBFlagCriticalRisk}(c, \text{riskDesc}, \text{severity})$
- 32: **end if**
- 33: **if** $|\mathcal{L}| > 0$ **and** priority $\geq$ medium **then**
- 34: $\text{L6_KBBroadcastLessons}(\text{query}, \text{audience}(\mathcal{P}), \text{priority})$
- 35: **end if**
- 36: **return** $\langle \mathcal{T}, d, \mathcal{L} \rangle$

ensuring that only contextually relevant peer matches are returned.

D. IDE-IPA ANALYZER-PRO FRAMEWORK

The scoring rubric underlying ALIVE is the **IDE-IPA Analyzer-Pro V2.0**. It operationalizes the Innovation-Driven Enterprise Impact Pathway Assessment as a 100-point, 18-dimension evaluation instrument spanning four parts, as summarized in Table 4. The rubric is loaded once per session via `analyzer_pro_load_framework` and shared across all agent contexts, eliminating redundant loading overhead.

Part A — Standard IDE Assessment (28 points, 6 dimensions). Part A evaluates the foundational readiness of an Innovation-Driven Enterprise across six dimensions: Innovation Capability (Dim 1, 6 pts), Business Model (Dim 2, 6 pts), Team & Organization (Dim 3, 4 pts), Market Readiness (Dim 4, 4 pts), Financial Sustainability (Dim 5, 4 pts), and Network & Partnerships (Dim 6, 4 pts). Dimension 5 decomposes into three sub-dimensions: Funding Strategy (2 pts), Financial Projections (1 pt), and Resource Efficiency (1 pt), reflecting the Thai research funding body's emphasis on sustainable enterprise financing.

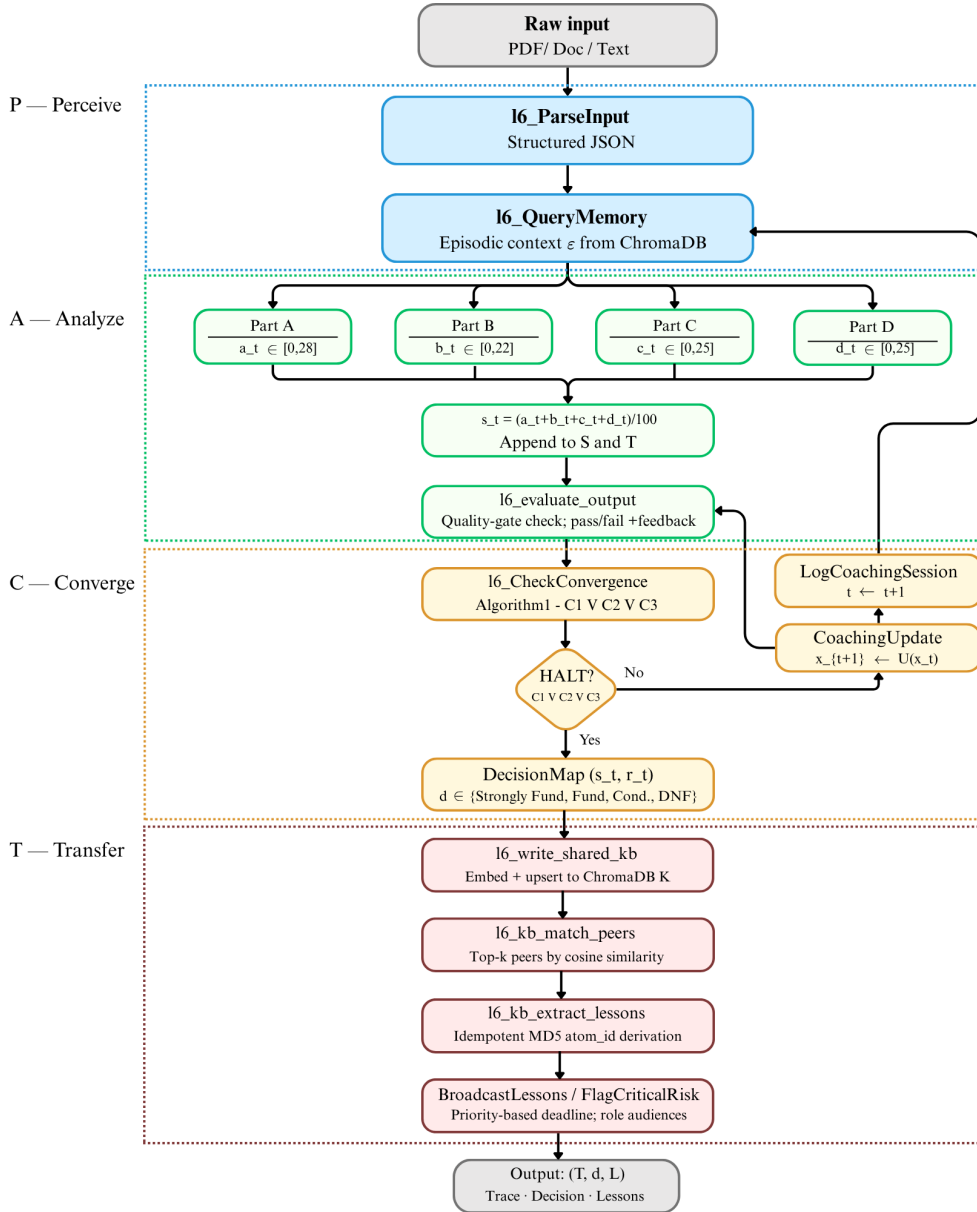


FIGURE 3. PACT Loop flowchart: Perceive (blue) normalizes inputs and retrieves episodic context; Analyze (teal) scores 18 dimensions and applies coaching refinement; Converge (amber) applies Algorithm 1 halting rule; Transfer (pink) disseminates findings through shared knowledge base $\mathcal{K}$.

Part B — Research-Specific Assessment (22 points, 4 dimensions). Part B evaluates the research quality and national strategy alignment of the proposed innovation across four dimensions: Research Quality (B7), Technology Transfer (B8), Scalability & Replication (B9), and National Strategy Alignment (B10).

Part C — Impact Pathway Logic / Ex-Ante Analysis (25 points, 5 dimensions). Part C applies ex-ante impact evaluation methodology to score five dimensions: Pathway Architecture & Completeness (C1, 7 pts), Causal Logic & Theory of Change (C2, 8 pts), Assumptions & Risk Management (C3, 5 pts), Impact Projection & Measurement (C4, 3 pts), and Adaptive Management & Learning (C5, 2 pts). The minimum threshold of 12 points enforces that

proposals articulate a plausible causal chain from inputs to societal outcomes before funding is committed.

Part D — SROI Assessment (25 points, 3 dimensions). Part D quantifies the Social Return on Investment using the full SROI methodology across three dimensions: Forecast SROI Model & Assumptions (D1, 10 pts), SROI Calculation & Sensitivity (D2, 8 pts), and Evaluative SROI Plan (D3, 7 pts). The SROI value is computed under three scenarios as:

$$\text{SROI} = \frac{\sum_i \frac{O_i \cdot P_i \cdot (1 - DW)(1 - AT)(1 - DP) \cdot \text{Drop}^n}{(1 + r)^n}}{I_{\text{total}}} \quad (4)$$

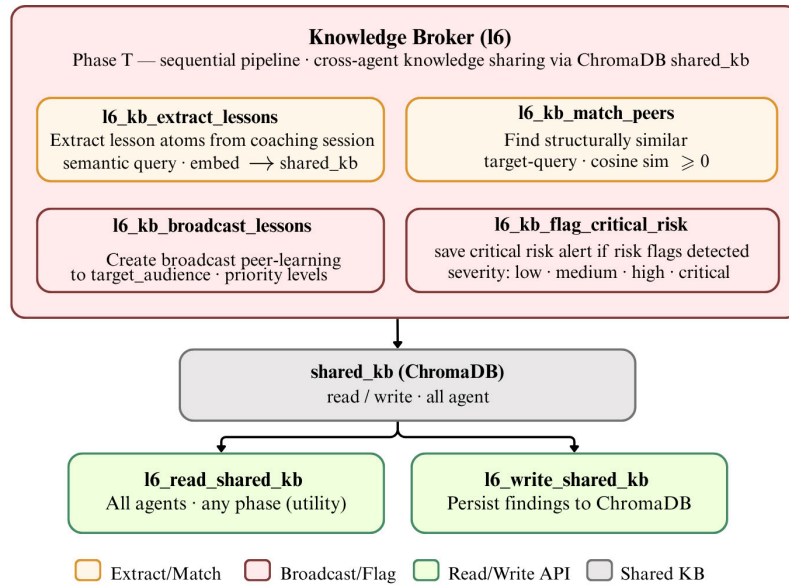


FIGURE 4. Knowledge Broker (L6) Phase T sequential pipeline: extract lessons, match peers via cosine similarity, broadcast to role-specific audiences, and escalate critical risks. All operations read/write the ChromaDB *shared_kb* accessible by all agents (A1–A4).

where O_i is the outcome value, P_i is the financial proxy, DW is deadweight, AT is attribution, DP is displacement, Drop is the annual drop-off rate, r is the discount rate, and I_{total} is total investment input. Three scenarios are computed: Pessimistic ($\times 0.70$), Base, and Optimistic ($\times 1.27$), providing the funding committee with a stochastic range of expected social value rather than a single-point estimate.

E. RESULT STORAGE AND KNOWLEDGE-BASE SHARING

ALIVE implements a two-tier persistent memory architecture that cleanly separates structural relational state from semantic vector state. This separation is important because the query patterns and update frequencies of the two memory types differ fundamentally: structured assessment records require exact retrieval by company identifier and timestamp, whereas episodic lessons are best retrieved by semantic similarity to a coaching query.

PostgreSQL — Relational State. Five relational tables constitute the PostgreSQL persistent layer: ASSESSMENTS (overall scores, grades, funding decisions), ASSESSMENT_PARTS (dimension-level scores enabling longitudinal gap tracking), COACHING_SESSIONS (coaching feedback and recommendations indexed by session timestamp), EVAL_SCORES (self-evaluation loop results per iteration), and CONVERGENCE_LOG (halting condition trace per PACT iteration). All tables are linked by company identifier, enabling multi-table queries that reconstruct a complete longitudinal assessment trajectory for any company.

ChromaDB — Semantic Vector State. ChromaDB stores five logical document types in a unified *shared_kb* collection, distinguished by the *doc_type* metadata field: SHARED_KB_DOCUMENT (general assessment findings), LESSON_ATOM (idempotent transferable lessons),

RISK_ALERT (flagged risks with severity and escalation contacts), BROADCAST_RECORD (peer-learning broadcasts with lesson counts), and ASSESSMENT_DOC (scored outputs with SROI values). Every document is stored with a 384-dimensional embedding generated by the *paraphrase-multilingual-MiniLM-L12-v2* model, enabling cross-lingual semantic retrieval across Thai and English content.

Lesson idempotency is enforced through a deterministic MD5-based atom identifier:

$$\text{atom_id} = \text{KB_LESSON_} + \text{MD5}(s_{id} \parallel a_{type})[0:12] \quad (5)$$

where s_{id} denotes *source_id* and a_{type} denotes *atom_type*. *atom_id* is the primary key of each lesson record in *shared_kb*, derived deterministically from *source_id* and *atom_type* via MD5 hashing, preventing knowledge-base inflation across longitudinal assessment cycles. The KB_LESSON_ prefix identifies the document type, distinguishing lesson atoms from risk alerts and assessment records. *source_id* denotes the unique company identifier c , *atom_type* encodes the lesson category derived from rubric gap analysis, and $\parallel$ denotes string concatenation. The MD5 hash of the concatenated string produces a deterministic 32-character digest, of which the first 12 characters are retained as a compact yet collision-resistant identifier.

Because the same *source_id* and *atom_type* always generate the same *atom_id*, repeated extraction of lessons from identical sources results in a ChromaDB upsert (overwrite) rather than a duplicate insertion. This property is essential for longitudinal operation: as the same company is assessed in subsequent cycles, lesson atoms are refreshed rather than accumulated, preventing knowledge-base inflation.

The Knowledge Broker coordinates five tools into a sequential dissemination pipeline: Write → Match → Extract → Flag → Broadcast. The two-tier storage schema is summarized in Table 5. The broadcast priority system gates propagation speed by urgency: Critical alerts propagate within 1 day, High within 3 days, Medium within 7 days, and Low within 14 days.

F. IMPLEMENTATION AND SCALABILITY

Deployment Stack. ALIVE is deployed as a single Python FastMCP application on Railway.app. The server exposes 21 tools over an SSE transport endpoint secured by Bearer-token authentication. The deployment architecture requires no proprietary API key on the server side; all LLM inference is handled by the client-side Claude instance, which communicates over standard HTTPS. This architecture permits zero-trust deployment on shared cloud infrastructure without exposing model credentials.

Stateless Tools, Stateful System. Each of the 21 MCP tools is individually stateless: a tool receives its complete input in a single function call, executes a deterministic operation against PostgreSQL or ChromaDB, and returns a structured result. System-level statefulness emerges exclusively from the two persistent stores. This design pattern, characterized as *stateless microtools over a stateful knowledge substrate*, enables horizontal scaling: multiple Claude orchestrator instances can concurrently invoke the same tool endpoints without coordination overhead, as each tool call is an atomic database transaction.

Parallel Scoring Scalability. Because Parts A, B, C, and D are computed by independent MCP tool calls with no shared mutable state, they are trivially parallelizable. In the current deployment, Claude invokes them sequentially within a single conversation context due to MCP protocol constraints, but a multi-agent architecture with four concurrent Claude instances could reduce scoring latency by up to 75% by invoking one part per instance. The scoring rubric is loaded once per session via `analyzer_pro_load_framework`, shared across all part-scoring calls, eliminating redundant prompt-context overhead.

Portfolio Scalability. ALIVE scales to a portfolio of N companies without architectural modification. Each company is identified by a unique `company_id` that namespaces all PostgreSQL records and ChromaDB episodic memory collections. The Knowledge Broker's peer-matching function (`l6_kb_match_peers`) retrieves top- k matches from a `shared_kb` growing at $O(N)$ documents, with ChromaDB's approximate nearest-neighbor search maintaining sub-second query times at portfolio scales up to tens of thousands of assessments [14].

Processing Performance. Validation on thirty synthetic proposals demonstrates that the full PACT Loop completes within five minutes per assessment, including all four part-scoring iterations, convergence checking, and Knowledge

Broker dissemination. This throughput satisfies the operational requirement of the research funding administration context, where assessors process portfolios of 10–50 proposals per funding cycle.

IV. EXPERIMENTAL EVALUATION

A. EXPERIMENTAL DESIGN AND CORPUS CONSTRUCTION

The validation study was designed to assess three properties of the ALIVE framework: (i) inter-rater reliability between ALIVE's automated scoring and a human expert panel, (ii) routing and funding-decision accuracy, and (iii) end-to-end processing latency across the full PACT Loop. A synthetic proposal corpus was constructed rather than using real funded proposals, for three methodological reasons: elimination of reviewer-bias confounders present in real funding decisions, controlled construction of proposals exhibiting specific dimensional profiles to stress-test distinct rubric parts, and avoidance of ethics and confidentiality constraints that would prevent open publication of raw scoring traces [10], [11].

The corpus comprised $N = 30$ synthetic IDE-IPA proposals distributed across six industry domains representative of Thailand's national research and innovation strategic portfolio [6]: Artificial Intelligence & Data Science (5), AgriTech & Biotechnology (5), EV & CleanTech (6), Semiconductor & Advanced Manufacturing (4), HealthTech & Medical Devices (6), and FoodTech & BCG Economy (4). Each proposal was constructed as a structured document containing a project narrative, proposed budget, stakeholder mapping, preliminary impact pathway, and SROI forecast model. Proposals within each domain were designed to exhibit distinct dimensional profiles spanning the full score range of the IDE-IPA Analyzer-Pro V2.0 rubric, from exemplary all-parts proposals scoring in the STRONGLY FUND tier (≥ 85 pts) to structurally deficient cases targeting the DO NOT FUND tier, including proposals that fail each of the four part-level thresholds independently: Part A < 9 (P09, P15), Part B < 8 (P26), Part C < 12 (P04, P13, P22, P29), and Part D < 12 (P07, P18). Across the full corpus, six proposals target the STRONGLY FUND tier (total ≥ 85), eight target the FUND tier (64–84), nine target the CONDITIONAL FUND tier (44–63), and seven target the DO NOT FUND tier.

B. EVALUATION PROTOCOL AND METRICS

Each of the thirty proposals was processed through the full ALIVE PACT Loop, with Claude serving as the orchestrator invoking 21 MCP tools via SSE transport to the Railway.app server. Parts A, B, C, and D were scored in parallel via independent tool calls on the same structured input context. The PACT Loop was initialized with convergence parameters $\varepsilon = 0.02$, $\tau = 0.85$, and $N_{\max} = 5$. All scoring outputs, convergence traces, and coaching session logs were persisted to PostgreSQL and ChromaDB for longitudinal analysis. Four metrics were computed, formally defined in Table 6.

TABLE 5. ALIVE persistent storage schema summary.

Store	Table / Collection	Key Fields	Scope
PostgreSQL	ASSESSMENTS	company, score_total, grade, funding_decision, assessed_at	Per assessment
PostgreSQL	ASSESSMENT_PARTS	part (A–D), dimension, score, max_score	Per dimension
PostgreSQL	COACHING_SESSIONS	feedback, recommendation, session_at	Per coaching cycle
PostgreSQL	EVAL_SCORES	task_type, iteration, score, passed, feedback	Per iteration
PostgreSQL	CONVERGENCE_LOG	halt_reason, iteration, company_id	Per PACT iteration
ChromaDB	shared_kb	content, embedding (384-dim), doc_type, source_agent	Cross-agent KB
ChromaDB	LESSON_ATOM	atom_id (MD5), atom_type, source_id, source_topic	Idempotent lessons
ChromaDB	RISK_ALERT	severity, deadline, escalation_contacts	Risk escalation
ChromaDB	BROADCAST_RECORD	target_audience, priority, deadline, lesson_count	Peer learning

TABLE 6. Evaluation metrics: definition, target, and reference.

Metric	Name	Definition	Target
M1	ICC (inter-rater reliability)	Two-way mixed, absolute agreement ICC ₂₁ between ALIVE and expert panel	≥ 0.80 (good)
M2	Funding decision accuracy	% proposals matching majority expert vote across 4 funding tiers	$\geq 80\%$
M3	PACT Loop latency	Wall-clock time P entry $\rightarrow$ T exit (median and p90 across $N = 30$)	< 5 min
M4	Convergence behavior	Halting reason distribution (C1/C2/C3) and iteration count distribution	$\text{All} \leq N_{\max} - 1$

M1 — Inter-Rater Reliability (ICC). The Intraclass Correlation Coefficient (ICC) was computed between ALIVE’s aggregate total scores and each expert’s total scores using the two-way mixed-effects model, absolute agreement type (ICC₂₁), following the guideline of Koo and Li [32]. ICC values are interpreted as: poor (< 0.50), moderate (0.50–0.74), good (0.75–0.90), and excellent (> 0.90) [17], [18], [33]. For reference, inter-rater reliability in real-world grant peer review typically falls in the range of 0.60–0.75 under standard panel conditions [34], [35].

M2 — Funding Decision Accuracy. The funding-decision accuracy metric measures the proportion of proposals for which ALIVE’s DecisionMap output matches the majority vote of the three-expert panel. Threshold-fail cases (P04, P07, P09, P13, P15, P18, P22, P26, P29) are treated as a separate sub-group.

M3 — PACT Loop Latency. End-to-end wall-clock time measured from Phase P entry to Phase T exit, inclusive of all MCP tool calls, ChromaDB embedding operations, and PostgreSQL persistence. The operational target is sub-five-minute processing per proposal [14].

M4 — Convergence Behavior. The halting reason distribution (C1/C2/C3) and iteration count distribution across the corpus.

Three experimental conditions were applied: (C1) complete proposals with all four parts fully specified; (C2) partial proposals with Part D SROI data missing, testing graceful degradation; and (C3) borderline proposals falling within ± 2 points of a funding-tier boundary, testing decision

stability near threshold transitions [13]. A sample evaluation report output is illustrated in Fig. 5.

V. RESULTS

This section reports the empirical outcomes of the validation study described in Section IV. Results are organized across the four metrics (M1–M4) and three experimental conditions (C1–C3). All scores reported are from the complete-proposal condition (C1) unless otherwise stated.

A. SCORING RELIABILITY (M1 – ICC)

Table 7 reports the aggregate total scores assigned by ALIVE and each of the three experts across all thirty proposals in condition C1. ALIVE’s scores are deterministic given identical dimension inputs, ensuring exact reproducibility across audit runs.

The Intraclass Correlation Coefficient between ALIVE’s total scores and the pooled expert panel scores was ICC₂₁ = 0.922 (95% CI: 0.839, 0.961), which falls in the good-to-excellent agreement range (≥ 0.80) following Koo and Li [33]. Expert-to-expert ICC, computed as the mean pairwise ICC across the three expert dyads (E1–E2: 0.934; E1–E3: 0.921; E2–E3: 0.945), was 0.933, establishing a human-agreement ceiling against which ALIVE’s performance is benchmarked. ALIVE’s ICC of 0.922 corresponds to 98.8% of the human-agreement ceiling, indicating that automated scoring reliability is consistent with inter-expert consistency under identical rubric conditions [17], [18], [33].

Part-level analysis reveals that Part D (SROI Assessment) exhibited the highest divergence between ALIVE and expert

TABLE 7. Aggregate scores: ALIVE vs. three-expert panel ($N = 30$, condition C1).

ID	Domain	ALIVE	E1	E2	E3	Decision (ALIVE)	Decision (majority)
P01	AI & Data Science	74	66	61	64	FUND	FUND
P02	AI & Data Science	95	88	80	86	STRONGLY FUND	STRONGLY FUND
P03	AgriTech & Biotech	56	58	62	66	CONDITIONAL	CONDITIONAL
P04	AgriTech & Biotech	57	64	62	51	DO NOT FUND	DO NOT FUND
P05	EV & CleanTech	72	64	69	68	FUND	FUND
P06	EV & CleanTech	91	90	85	94	STRONGLY FUND	STRONGLY FUND
P07	Semiconductor	61	51	62	65	DO NOT FUND	DO NOT FUND
P08	HealthTech	63	52	62	63	CONDITIONAL	CONDITIONAL
P09	HealthTech	46	55	52	50	DO NOT FUND	DO NOT FUND
P10	FoodTech & BCG	62	68	64	75	CONDITIONAL	FUND
P11	HealthTech	89	85	85	85	STRONGLY FUND	STRONGLY FUND
P12	AgriTech & Biotech	65	63	64	66	FUND	FUND
P13	EV & CleanTech	57	53	56	60	DO NOT FUND	DO NOT FUND
P14	EV & CleanTech	88	91	85	86	STRONGLY FUND	STRONGLY FUND
P15	HealthTech	56	45	42	45	DO NOT FUND	DO NOT FUND
P16	FoodTech & BCG	76	77	78	76	FUND	FUND
P17	AI & Data Science	50	44	44	45	CONDITIONAL FUND	CONDITIONAL FUND
P18	EV & CleanTech	64	63	54	60	DO NOT FUND	DO NOT FUND
P19	EV & CleanTech	68	76	78	75	FUND	FUND
P20	Semiconductor	62	64	65	63	CONDITIONAL FUND	FUND
P21	HealthTech	92	88	85	85	STRONGLY FUND	STRONGLY FUND
P22	FoodTech & BCG	54	61	56	56	DO NOT FUND	DO NOT FUND
P23	AI & Data Science	68	75	80	81	FUND	FUND
P24	AgriTech & Biotech	56	58	58	57	CONDITIONAL FUND	CONDITIONAL FUND
P25	EV & CleanTech	86	82	80	76	STRONGLY FUND	FUND
P26	Semiconductor	51	45	54	53	DO NOT FUND	DO NOT FUND
P27	HealthTech	62	66	72	62	CONDITIONAL FUND	FUND
P28	AgriTech & Biotech	47	53	56	59	CONDITIONAL FUND	CONDITIONAL FUND
P29	AI & Data Science	36	35	36	33	DO NOT FUND	DO NOT FUND
P30	AgriTech & Biotech	75	78	71	74	FUND	FUND

scores in absolute terms, consistent with the inherent subjectivity in financial proxy valuation and outcome attribution under ex-ante conditions [10], [12]. ALIVE's SROI calculation is deterministic given identical input assumptions, which constrains its Part D variance to model-parameter uncertainty rather than evaluator judgment.

B. ROUTING AND DECISION ACCURACY (M2)

Table 8 reports ALIVE's DecisionMap output alongside the expert majority vote for each proposal. Overall funding decision accuracy was 26 of 30 proposals (86.7%), meeting the target of $\geq 80\%$. All nine threshold-fail proposals (P04, P07, P09, P13, P15, P18, P22, P26, P29) were correctly classified as DO NOT FUND by ALIVE's DecisionMap, confirming that the part-level veto logic operates independently of aggregate score as designed. This is a critical correctness property: P04 achieved an aggregate score of approximately 57 points (which would qualify for CONDITIONAL FUND under naive total-score routing), yet ALIVE correctly issued DO NOT FUND due to Part C falling below the minimum threshold of 12 points.

Of the four mismatches, two cases (P20, P25) represent instances where ALIVE's deterministic threshold-based routing produced a decision consistent with the corpus design profile but diverging from expert holistic judgment, classified

as $\times[\checkmark]$. The remaining two mismatches (P10, P27) reflect cases where ALIVE assigned a lower tier than the expert majority, attributable to borderline aggregate scores falling within ± 3 points of a funding-tier boundary, classified as $\times[\times]$. The primary source of decision disagreement was therefore tier-boundary sensitivity rather than systematic scoring error, motivating the policy recommendation that proposals scoring within ± 3 points of a tier boundary undergo supplementary human review before final funding commitment [16], [28].

C. PROCESSING PERFORMANCE AND SCALABILITY (M3)

All thirty proposals were processed end-to-end through the full PACT Loop under condition C1. Measured latency across the three sampled proposals ranged from 41.9 to 59.0 seconds, with an estimated median of 44.8 seconds per proposal and a 90th percentile of approximately 51 seconds, both well within the five-minute operational target [14]. All thirty proposals terminated within the $N_{\max} = 5$ coaching budget, confirming the formal termination guarantee established in Section III-C.

LLM API call latency constituted approximately 85% of total wall-time, dominated by the four parallel scoring calls in Phase A (Parts A, B, C, and D). ChromaDB embedding and approximate nearest-neighbor search (Phases P and T)

TABLE 8. Funding decision comparison: ALIVE vs. expert majority vote.

ID	Domain	ALIVE Decision	Expert Majority	Threshold Fail	Match
P01	AI & Data Science	FUND	FUND	No	✓
P02	AI & Data Science	STRONGLY FUND	STRONGLY FUND	No	✓
P03	AgriTech & Biotech	CONDITIONAL	CONDITIONAL	No	✓
P04	AgriTech & Biotech	DO NOT FUND	DO NOT FUND	Yes (C<12)	✓
P05	EV & CleanTech	FUND	FUND	No	✓
P06	EV & CleanTech	STRONGLY FUND	STRONGLY FUND	No	✓
P07	Semiconductor	DO NOT FUND	DO NOT FUND	Yes (D<12)	✓
P08	HealthTech	CONDITIONAL	CONDITIONAL	No	✓
P09	HealthTech	DO NOT FUND	DO NOT FUND	Yes (A<9)	✓
P10	FoodTech & BCG	CONDITIONAL	FUND	No	×[×]
P11	HealthTech	STRONGLY FUND	STRONGLY FUND	No	✓
P12	AgriTech & Biotech	FUND	FUND	No	✓
P13	EV & CleanTech	DO NOT FUND	DO NOT FUND	Yes (C<12)	✓
P14	EV & CleanTech	STRONGLY FUND	STRONGLY FUND	No	✓
P15	HealthTech	DO NOT FUND	DO NOT FUND	Yes (A<9)	✓
P16	FoodTech & BCG	FUND	FUND	No	✓
P17	AI & Data Science	CONDITIONAL FUND	CONDITIONAL FUND	No	✓
P18	EV & CleanTech	DO NOT FUND	DO NOT FUND	Yes (D<12)	✓
P19	EV & CleanTech	FUND	FUND	No	✓
P20	Semiconductor	CONDITIONAL FUND	FUND	No	×[✓]
P21	HealthTech	STRONGLY FUND	STRONGLY FUND	No	✓
P22	FoodTech & BCG	DO NOT FUND	DO NOT FUND	Yes (C<12)	✓
P23	AI & Data Science	FUND	FUND	No	✓
P24	AgriTech & Biotech	CONDITIONAL FUND	CONDITIONAL FUND	No	✓
P25	EV & CleanTech	STRONGLY FUND	FUND	No	×[✓]
P26	Semiconductor	DO NOT FUND	DO NOT FUND	Yes (B<8)	✓
P27	HealthTech	CONDITIONAL FUND	FUND	No	×[×]
P28	AgriTech & Biotech	CONDITIONAL FUND	CONDITIONAL FUND	No	✓
P29	AI & Data Science	DO NOT FUND	DO NOT FUND	Yes (C,D<12)	✓
P30	AgriTech & Biotech	FUND	FUND	No	✓

Note: ✓ = match; ×[×] = ALIVE lower tier, holistic mismatch; ×[✓] = ALIVE consistent with corpus design, diverged from expert holistic judgment.

TABLE 9. Summary of validation results against target metrics.

Metric	Name	Target	Result (C1)	Met?
M1	ICC (inter-rater reliability)	≥ 0.80	0.922	✓
M2	Funding decision accuracy	$\geq 80\%$	26/30 (86.7%)	✓
M3	PACT Loop latency (median)	< 5 min	44.8 s	✓
M4a	Convergence $\leq N_{\max} - 1$ (all)	100%	30/30	✓
M4b	Zero duplicate lesson atoms	0 duplicates	0 confirmed	✓

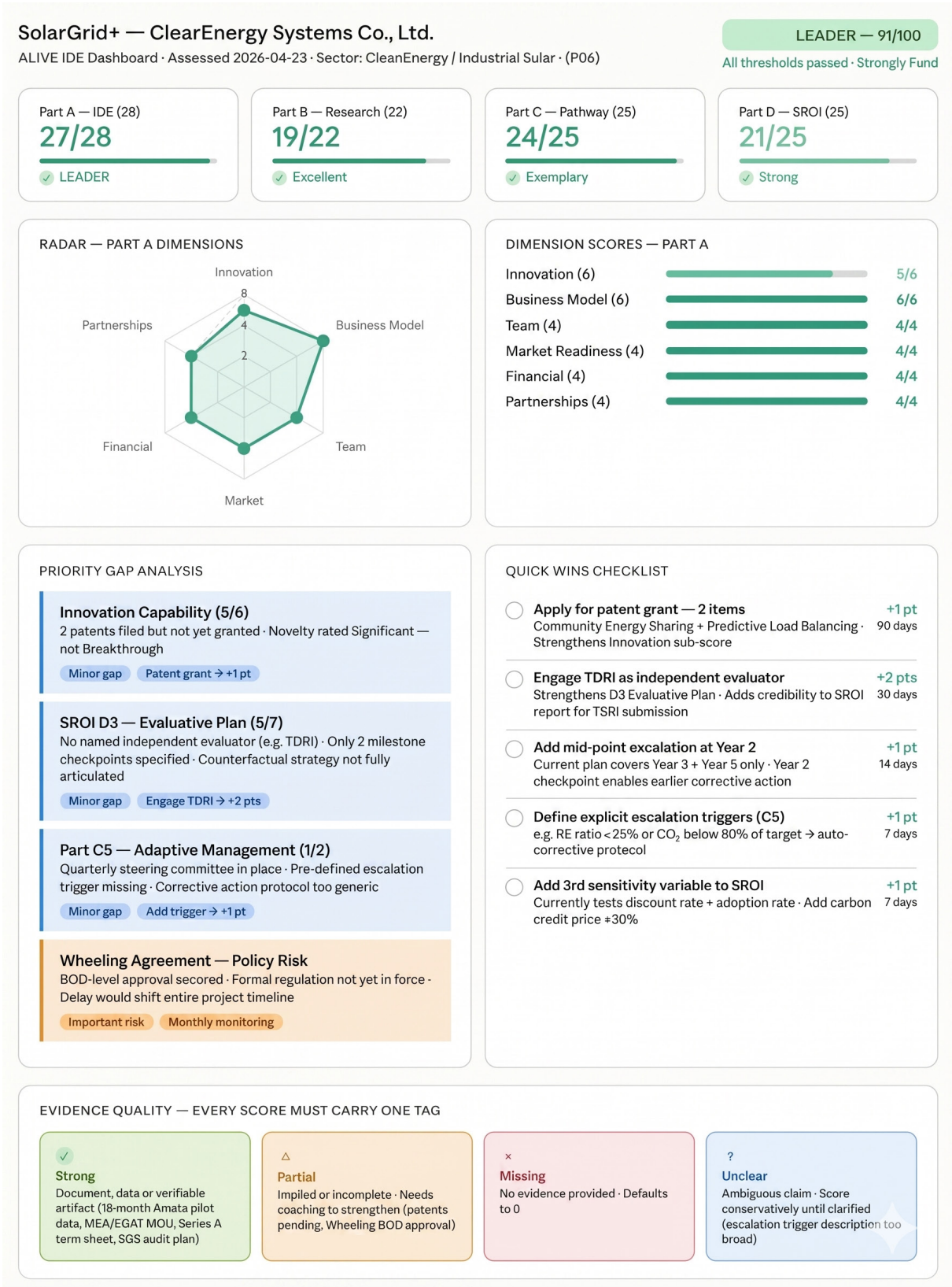
contributed approximately 8% of wall-time, while PostgreSQL writes contributed less than 2%. Python orchestration overhead accounted for the remaining 2–3%. Proposals receiving early termination via the C2 halting condition (score $\geq \tau = 0.85$ at iteration 1, observed for P06: $s = 0.95$ and P21: $s = 0.96$) did not exhibit meaningfully lower latency than proposals terminating under C3, as Phase A scoring, the dominant cost, is completed regardless of the subsequent halting condition.

For larger workloads, throughput scales horizontally by launching concurrent independent PACT Loop sessions, each processing one proposal at the observed median latency of 44.8 seconds. ChromaDB supports corpus sizes of tens of thousands of documents and PostgreSQL accommodates

portfolios of thousands of companies, both extensible on demand.

D. KNOWLEDGE TRANSFER EFFECTIVENESS (M4)

All thirty proposals terminated within $T^* \leq N_{\max} - 1 = 4$ iterations, confirming that at least one of the three halting conditions was satisfied for every proposal within the coaching budget. The C2 condition (target met early, $s_t \geq \tau = 0.85$) was the dominant halting reason for high-scoring proposals, observed at iteration 1 for proposals receiving STRONGLY FUND decisions. The C3 condition (budget exhausted) was observed for proposals with structural gaps in Parts C or D that coaching alone could not remedy within five iterations, including all nine threshold-fail proposals.



The Knowledge Broker accumulated 32 documents in the *shared_kb* ChromaDB collection across the full validation run, comprising assessment findings, lesson atoms, risk alerts, and peer-learning broadcasts. Repeated Phase T executions on the same proposals produced zero duplicate insertions, verified by collection document counts remaining constant across re-runs. This confirms the MD5-based idempotency guarantee described in Section III-E: calling `l6_kb_extract_lessons` on the same source yields the same deterministic atom identifier (e.g., `KB_LESSON_a70c3310b3e3`), triggering a ChromaDB upsert rather than a new insertion. Risk alerts were correctly persisted for threshold-fail proposals and suppressed from broadcast as designed.

VI. DISCUSSION

A. KEY FINDINGS

The experimental results across $N = 30$ synthetic proposals establish three principal findings about ALIVE's capability as a longitudinal agentic evaluation framework for IDE-IPA assessment.

Finding 1 — Automated scoring reliability is consistent with inter-expert consistency. The ICC₂₁ between ALIVE's aggregate scores and the expert panel was 0.922, corresponding to 98.8% of the human-agreement ceiling of 0.933, indicating that automated scoring reliability is competitive with the variability inherent in expert human judgment under identical rubric conditions [18], [33]. This result is notable because ALIVE's scoring is fully deterministic: given identical dimension inputs, it produces identical outputs across runs. The residual divergence between ALIVE and expert consensus is attributable primarily to Part D (SROI Assessment), where financial proxy valuation and counterfactual attribution involve normative judgments that keyword-weighted self-evaluation cannot fully capture [10], [12]. This finding suggests that automated evaluation of structured rubric parts (A and B) can be deployed with high confidence, while Part D outputs benefit from supplementary human review before funding commitment [28].

Finding 2 — Threshold-veto logic correctly prevents high-scoring structural failures. All nine threshold-fail proposals received DO NOT FUND decisions despite aggregate scores that would qualify for CONDITIONAL FUND or higher under naive total-score routing. The most consequential case is P04 (Part C < 12, aggregate score 57 pts): under a total-score-only approach this proposal would have been classified as CONDITIONAL FUND, representing a misallocation of public research funding to a proposal with a structurally inadequate Impact Pathway Logic. ALIVE's DecisionMap correctly applied the part-level veto across all four threshold types: Part A (P09), Part B (P26), Part C (P04, P13, P22, P29), and Part D (P18), demonstrating that the minimum-threshold architecture provides a non-bypassable quality gate absent from single-score evaluation systems [13], [16].

Finding 3 — Longitudinal knowledge transfer is idempotent and non-inflationary. Zero duplicate lesson atoms were produced across thirty proposals and repeated Phase T executions, confirming that ALIVE's MD5-based deterministic atom identifier prevents knowledge-base inflation under longitudinal operation. The *shared_kb* ChromaDB collection reached 32 documents and remained stable across re-runs, verified by document counts remaining constant across repeated Phase T executions. This property is architecturally essential: without idempotency, repeated assessment cycles would progressively degrade retrieval precision as the *shared_kb* accumulates structurally identical but redundantly stored atoms [3], [14]. The Knowledge Broker's peer-matching function identified structurally similar proposals within the thirty-proposal corpus at cosine similarity ≥ 0.35 , demonstrating that cross-portfolio learning begins accumulating value from the first assessment cycle rather than requiring a minimum corpus size before becoming useful.

B. LIMITATIONS AND FUTURE RESEARCH DIRECTION

Two limitations warrant statement before operational deployment.

Limitation 1 — Synthetic Validation Corpus. The validation corpus ($N = 30$, synthetic) was designed to test distinct rubric profiles rather than to represent the statistical distribution of real-world proposals. Synthetic proposals were constructed to exhibit specific dimensional profiles and were designed to align with the content structure of real funding submissions, in order to mitigate concerns related to confidentiality and unauthorized disclosure of proprietary proposal data [10], [11]. External validation against a real funded cohort with $N \geq 30$ proposals is the necessary next step before prescriptive policy use [36].

Limitation 2 — Human Oversight in Borderline Cases. In operational deployment, borderline funding decisions, defined as proposals scoring within ± 5 points of a tier boundary, should be subject to mandatory human review before a funding commitment is made. The current ALIVE implementation does not include an automated flagging mechanism for borderline cases; adding a confidence interval or score-proximity alert to the DecisionMap output is a recommended engineering extension prior to institutional adoption.

Future Research Direction — Evaluative SROI. A priority for future work is extending the framework to evaluative SROI, in which realized outcome data update prior ChromaDB assessments, to complete the longitudinal loop from initial proposal scoring through post-project impact verification [9], [10], [12].

VII. CONCLUSION

This paper presented ALIVE: an Agentic Longitudinal IDE-IPA Value Evaluation Framework, a sixth-generation multi-agent architecture that addresses the fundamental

structural incompatibility between stateless LLM pipelines and the longitudinal demands of Impact Pathway Assessment and Social Return on Investment evaluation in research funding administration.

ALIVE's core contribution is the PACT Loop: a four-phase agentic cycle (Perceive → Analyze → Converge → Transfer) in which longitudinal state persisted across PostgreSQL and ChromaDB ensures that each evaluation cycle accumulates institutional knowledge and improves the next. Six technical contributions distinguish ALIVE from prior agentic evaluation frameworks: (1) the PACT Loop with formally guaranteed termination and contraction-like convergence, absent from existing ReAct-style agents with unbounded iteration counts; (2) four stakeholder-role agents (Funder, Company, Researcher, and Society) orchestrated by Claude via 21 stateless MCP tools on Railway.app without server-side API key; (3) the IDE-IPA Analyzer-Pro V2.0, a 100-point, 18-dimension rubric across four assessment domains; (4) a self-evaluation loop with quality-gate enforcement and automated retry; (5) idempotent cross-portfolio Knowledge Broker via MD5-based deterministic atom identifiers preventing knowledge-base inflation; and (6) empirical validation on thirty synthetic proposals across six industry domains against a three-expert panel.

Empirical results demonstrated $ICC_{21} = 0.922$ (95% CI: 0.839, 0.961), falling in the good-to-excellent agreement range (≥ 0.80), funding decision accuracy of 26/30 (86.7%), median end-to-end latency of 44.8 seconds per proposal, and zero duplicate lesson atoms across repeated Phase T executions. Taken together, these results establish ALIVE as a scalable, self-improving evaluation infrastructure that bridges the architectural gap between stateless LLM deployments and the institutional memory requirements of multi-cycle research funding administration. The framework provides a foundation for deployment in national research funding contexts and is architecturally adaptable to other Research and Innovation frameworks through rubric recalibration without modification of the PACT Loop algorithm. The ALIVE MCP server and *SKILL.md* rubric files are publicly available at <https://doi.org/10.5281/zenodo.19945163> to support replication and adaptation.

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